llmXive Automated Review of Guava: An Effective and Universal Harness for Embodied Manipulation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (8 lenses): Claim Accuracy, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The manuscript generally supports its claims with experimental data, but there are several instances where citations and figure references do not align with the attributed claims, and specific numerical claims lack direct tabular support.

First, in Section 3 ("Designing Effective Harness"), the text states: "Figure~\ref{fig:harness} validates our design choices by demonstrating the performance of \texttt{GPT-5.4}...". However, the label `\label{fig:harness}` is assigned to the ablation study figure (the three subplots), while the text refers to the "Teaser" image as `fig:teaser` in the LaTeX source (though the label in the code is `fig:teaser` for the first figure and `fig:harness` for the second). The text explicitly says "Figure~\ref{fig:harness} validates...", which is correct for the ablation figure, but the sentence structure implies the *teaser* figure might be the one validating the specific GPT-5.4 performance if the reader confuses the labels. More critically, the text claims the figure demonstrates performance of "GPT-5.4", but the figure caption and the text in Section 4 clarify that the ablation study compares different *configurations* (iterative vs one-shot, etc.) often using a frontier model, but the specific model name "GPT-5.4" is not explicitly defined as the *only* model in that specific figure's caption in the provided text, though it is implied. The primary issue is the potential confusion between the Teaser (Fig 1) and the Ablation (Fig 2) regarding which one "validates" the design choices. The text says "Figure~\ref{fig:harness} validates...", which is correct, but the previous sentence mentions "Figure~\ref{fig:teaser}" (implied by context of "Guava Overview"). The text flow is slightly confusing but the citation to `fig:harness` for the ablation is technically correct. However, the text says "Figure~\ref{fig:harness} validates... where multimodal setting... demonstrate a consistent higher performance". The figure caption supports this. The error is likely in the *naming* of the model in the text if the figure uses a different baseline, but the text says "GPT-5.4" which is the baseline used in Table 1.

A more significant issue is in Section 4, "Finding 2". The text claims: "On OOD tasks, \texttt{Guava-Agent-4B} maintains strong performance, achieving 100% success on \textit{move object away} and 90% on \textit{set table}." The task "move object away" does not appear in Table 1 (Simulation) or the list of tasks in Section 4.1. While it might be in the real-world evaluation (Figure 3), the specific claim of "100%" for a task not listed in the main results table makes it difficult to verify the accuracy of the claim without the figure's internal data. The text should either list the task in the table or clarify that this specific task was only evaluated in the real-world subset and provide the exact count (e.g., "10/10").

Furthermore, in the Appendix under "Additional Results", the text cites `Gemini-3.1-Pro` with `\citep{fu2026capxframeworkbenchmarkingimproving}` (the CaP-X paper) and `Claude-Sonnet-4.6` with `\citep{qwen2026qwen35}` (the Qwen paper). These are clear citation errors where the model name is attributed to a paper that does not introduce or primarily feature that model. This misrepresents the source of the model capabilities.

Finally, the claim of "fewer than 2K trajectories" is supported by the appendix count of 1,934, but scientific writing typically prefers the exact number in the main text to avoid ambiguity.

These issues are primarily writing and citation accuracy errors that do not invalidate

the core science but require correction to ensure the claims are precisely supported by the provided evidence.
Action items.

- [writing] In Section 3 (Method), the text claims Figure 1 validates design choices using 'GPT-5.4'. However, Figure 1 is the 'Teaser' image, while Figure 2 (labeled fig:harness) contains the ablation study. The text incorrectly attributes the validation data to the wrong figure.
- [writing] In Section 4 (Results), Finding 2 claims Guava-Agent-4B achieves '100% success on move object away' in real-world OOD tasks. However, Table 1 (sim_baseline) does not list 'move object away' as a task, and the real-world results are only presented in Figure 3 without a corresponding data table. The specific 100% claim lacks a verifiable source in the provided text or tables.
- [writing] In Appendix 'Additional Results', the text cites 'Gemini-3.1-Pro' with reference [fu2026capxframeworkbenchmarkingimproving] (CaP-X paper) and 'Claude-Sonnet-4.6' with reference [qwen2026qwen35] (Qwen paper). These citations are factually incorrect as they attribute model names to unrelated papers.
- [writing] In Section 3, the text states 'fewer than 2K trajectories' were used. Appendix 'Summary' specifies '1,934 trajectories'. While 1,934 is fewer than 2,000, the phrasing 'fewer than 2K' is slightly ambiguous in a scientific context where exact counts are preferred. Consider stating the exact number in the main text for precision.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on field-specific acronyms and shorthand that are not defined at their first occurrence, creating barriers for non-specialist readers. In Section 1, the term "ReAct" is introduced as a proper noun for a workflow without explicitly stating it stands for "Reasoning and Acting" or briefly explaining the mechanism, assuming the reader already knows the specific 2023 paper's terminology. Similarly, "VLM" is used immediately in the first paragraph without spelling out "vision-language model."

In Section 3.2, the training pipeline description introduces "SFT" and "GRPO" without definition. While standard in reinforcement learning literature, these acronyms obscure the methodology for a broader audience. The text should spell out "Supervised Fine-Tuning" and "Group Relative Policy Optimization" upon first use.

Furthermore, Section 4.1 categorizes tasks using "ID" and "OOD" without defining them as "in-distribution" and "out-of-distribution." This pattern of using acronyms without introduction is repeated in the Appendix with "SAM3," where the specific model name is

not explicitly linked to the acronym for the uninitiated reader. To align with the paper's claim of being a "universal" interface, the text should adopt a more inclusive style by defining these terms or using plain English equivalents where possible.
Action items.

- [writing] Define 'ReAct' at first use in Section 1 (Introduction). The term is used as a proper noun for a specific workflow pattern without explaining it stands for 'Reasoning and Acting' or citing the specific mechanism for non-specialists.
- [writing] Replace the acronym 'VLM' with 'vision-language model' at its first occurrence in Section 1. While common in the field, the paper claims to be 'universal' and should define standard acronyms for broader accessibility.
- [writing] Define 'SFT' (Supervised Fine-Tuning) and 'GRPO' (Group Relative Policy Optimization) at their first mention in Section 3.2. These are specific algorithmic terms that may be opaque to readers outside the immediate RL/LLM sub-community.
- [writing] Clarify 'SAM3' in the Appendix. The text mentions 'SAM3 segmentations' and cites a 2025 paper, but does not explicitly state that this refers to the 'Segment Anything Model' (or a specific version thereof) for readers unfamiliar with the specific model iteration.
- [writing] Replace 'OOD' with 'out-of-distribution' at first use in Section 4.1. The acronym is used repeatedly to categorize tasks without an initial definition, which hinders readability for general robotics or AI audiences.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The manuscript presents a coherent argument for the Guava harness framework, linking design principles to experimental results. However, there are critical logical inconsistencies in the reporting of quantitative results in Section 4 that undermine the validity of the conclusions drawn there.

Specifically, in Section 4, under "Finding 3: RL post-training substantially improves long-horizon reasoning...", the text explicitly states that RL post-training improves performance on the "shell game" task from 6.7% to 60.0% and on "place all red objects in basket" from 0.0% to 93.3%. These numbers are presented as the performance of the final "Guava-Agent-4B" model. However, Table 1 ("Quantitative comparison on simulation tasks") lists the "Guava-Agent-4B" success rate for "shell game" as 6.7% and for "place all red objects in basket" as 0.0%.

This creates a direct contradiction: the text claims the final model achieved 60.0% and

93.3% on these tasks, while the table (the primary source of truth for the results) reports 6.7% and 0.0%. It is unclear if the text is referring to a specific subset of runs, a different model variant not listed in the table, or if the table values are incorrect. As written, the conclusion that “Guava-Agent-4B” achieves these high scores on these specific tasks is unsupported by the provided data table.

Additionally, the claim in Section 3 that the model is trained on “fewer than 2K trajectories collected entirely in simulation” requires clarification regarding the nature of the “recovery trajectories.” The Appendix reveals that 38% of the data (743 trajectories) were generated by “manually adding error perturbations” to successful trajectories. While this is a valid data augmentation strategy, the phrasing “collected entirely in simulation” might imply that all 1,934 trajectories were raw, unmodified execution logs. Clarifying that a significant portion was synthetically perturbed would ensure the logical consistency of the data collection description.

Action items.

- [science] In Section 4, Finding 3 claims RL improves ‘shell game’ from 6.7% to 60.0% and ‘place all red objects’ from 0.0% to 93.3%. However, Table 1 lists the final Guava-Agent-4B scores for these tasks as 6.7% and 0.0% respectively. This is a direct contradiction between the text and the results table.
- [science] In Section 4, Finding 3 claims RL improves ‘place all red objects in basket’ from 0.0% to 93.3%. However, Table 1 lists the final Guava-Agent-4B score for this task as 0.0%. This contradicts the claim that the final model achieved 93.3% on this task.
- [writing] Section 3 claims training uses ‘fewer than 2K trajectories collected entirely in simulation’. The Appendix states 38% are ‘recovery trajectories’ generated by ‘manually adding error perturbations’. Clarify if these synthetically perturbed trajectories count as ‘collected’ data to ensure the claim is not misleading.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims that extend slightly beyond the strict boundaries of the provided experimental evidence, primarily regarding the scope of “universality” and the robustness of the “emergent capabilities.”

First, the title and abstract describe Guava as a “Universal Harness.” However, the experimental validation is confined to a single robotic embodiment (Franka Research 3) and a specific set of semantic tools (grasp, align, move, etc.). The paper explicitly acknowledges in the Limitations section that it “cannot handle dexterous manipulation” and assumes a “single-view image.” While the *principles* of the harness may be universal, the current instantiation and evaluation are not. The term "Universal" in the title is an overreach given the specific hardware and action space constraints. It would be more accurate to describe it as a "General" or "Effective" harness for the evaluated class of manipulation tasks.

Second, the abstract and introduction claim the model exhibits "strong emergent embodied capabilities" and "robust recovery" with minimal data. While the overall success rate (75.6%) is impressive, the results in Table 1 reveal significant fragility on specific long-horizon tasks. The model achieves 0.0% success on "place all red objects in basket" and only 6.7% on "shell game." Describing the capabilities as "strong" and "robust" without immediately qualifying these specific failure modes creates a skewed impression of the system's reliability. The claim of "strong generalization" is also challenged by the 0% performance on certain OOD tasks, suggesting the generalization is not uniform across all task types.

Finally, the claim of "zero-shot" transfer from simulation to the real world (Section 4, Finding 2) requires nuance. The policy weights are indeed not updated on real-world data. However, the "harness" itself includes specific perception modules (SAM3) and low-level controllers that are tuned or defined for the specific real-world hardware. The transfer is effectively from simulation to a *specific* real-world configuration, not to arbitrary real-world robots. The current phrasing risks implying a level of hardware agnosticism that the results do not support.

These are primarily issues of phrasing and qualification rather than fundamental scientific flaws. The data supports the conclusion that Guava is effective for the tested scenarios, but the language used to describe its scope and robustness is slightly hyperbolic.
Action items.

- [writing] The title and abstract claim 'Universal' applicability, but evaluation is limited to a single robot arm and specific primitives. Qualify 'universal' to 'general' or explicitly acknowledge the single-arm, non-dexterous limitation in the abstract.
- [writing] Claims of 'strong emergent capabilities' and 'robust recovery' are overreaching given 0% success on 'place all red objects in basket' and 6.7% on 'shell game' (Table 1). Contextualize 'strong' against these specific failure modes to avoid overgeneralizing robustness.
- [writing] The 'zero-shot' sim-to-real claim is misleading; transfer relies on specific hardware (Franka, RealSense) and tools (SAM3) defined in the harness. Clarify that the policy transfers to a specific embodiment, not arbitrary robots, to prevent overclaiming hardware agnosticism.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a framework for embodied manipulation using a harness and distilled models. From a safety and ethics perspective, the work is generally sound as it operates primarily in simulation (Robosuite) and uses a standard robotic arm (Franka Research 3) in controlled real-world settings. However, there are minor omissions regarding ethical oversight and safety protocols that require clarification.

First, the data curation process described in Appendix A.2 mentions “manual inspection on a subset of trajectories” to remove low-quality samples. The paper does not state whether this human-in-the-loop process was conducted under an Institutional Review Board (IRB) protocol or if the annotators were compensated and informed of the task nature. While the data is synthetic, the human labor involved in curation should be ethically accounted for, especially if the paper is intended for publication in venues requiring such disclosures.

Second, the reliance on SAM3 (Appendix A.2) for segmentation implies the processing of visual data. While the current experiments focus on objects, the authors should explicitly state that no human subjects were present in the real-world evaluation scenes (Sec 4.1) or, if they were, that appropriate privacy measures (e.g., blurring, consent) were taken. This is a standard requirement for robotics papers involving real-world deployment to prevent inadvertent surveillance or privacy violations.

Finally, the “Recovery Behavior Generation” (Appendix A.3) involves simulating failures like dropped objects. In the real-world experiments (Sec 4.1), the authors claim the agent can recover from “joint limits and unreachable poses.” The paper should briefly mention the physical safety measures (e.g., emergency stops, force limits, human exclusion zones) employed during these real-world trials to ensure that the agent’s recovery attempts do not cause damage to the hardware or injury to nearby personnel. Given the “zero-shot” nature of the transfer, the unpredictability of the agent’s behavior in the real world necessitates a clear statement on safety containment.

These points are minor and can be addressed with a few sentences in the Limitations or Ethics statement section, but they are necessary for a complete safety review.
Action items.

- [writing] The manuscript presents a framework for embodied manipulation using a harness and distilled models. From a safety and ethics perspective, the work is generally sound as it operates primarily in simulation (Robosuite) and uses a standard robotic arm (Franka Research 3) in controlled real-world settings. However, there are minor omissions regarding ethical oversight and safety protocols that require clarification. First, the data curation process described in Appendix A.2 mentions “manual inspecti

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The manuscript presents a compelling framework, but the scientific evidence supporting the central claims of universality and robustness is currently underpowered. While the reported success rates are promising, the sample sizes (N=15 for simulation, N=10 for real-world) are insufficient to draw statistically significant conclusions, particularly for the lower-performing tasks where confidence intervals likely overlap.

Specifically, the claim that the harness design is "universal" (Abstract) is tested only on GPT-5.4 in the ablation study (Fig 2). Without replicating these ablations on a second, distinct model family, it is unclear if the observed benefits are due to the harness or specific capabilities of the GPT-5.4 model. Similarly, the assertion that RL post-training "substantially improves" long-horizon reasoning (Finding 3) is based on only two tasks. This small N makes the result vulnerable to task-specific idiosyncrasies rather than a generalizable improvement in reasoning.

Furthermore, the real-world evaluation (N=10) is statistically weak for robotics benchmarks where variance is high. The difference between 60% and 40% success rates, for instance, is not statistically distinguishable with such a small sample. The authors should report confidence intervals (e.g., Wilson score intervals) for all success rates and increase the number of real-world trials to ensure the reported "strong generalization" is not a statistical artifact. Finally, the "fewer than 2K trajectories" claim (Abstract) should be contextualized with a power analysis or a discussion on the minimum sample size required to detect the observed effect sizes.

Action items.

- [science] The claim of 'fewer than 2K trajectories' (Abstract, Sec 1) lacks statistical power analysis. With N=15 episodes per task (Table 1), the confidence intervals for success rates (e.g., 6.7% vs 53.3%) are extremely wide. Report 95% CIs or standard errors for all success rates to validate the significance of the reported gains.
- [science] The RL ablation (Fig 4, Sec 4) compares SFT vs. RL on only two tasks ('shell game', 'place all red objects'). This sample size is insufficient to support the broad claim that RL 'substantially improves long-horizon reasoning' generally. Include results on at least 2-3 additional long-horizon tasks to rule out task-specific overfitting.
- [science] The harness ablation study (Fig 2) relies on a single frontier model (GPT-5.4). To support the claim that the design principles are 'universal' (Abstract), the ablation must be replicated on at least one other distinct model architecture (e.g., a different parameter scale or family) to demonstrate robustness against model-specific biases.
- [science] The Sim-to-Real transfer claim (Sec 4, Finding 2) is based on 10 episodes per task. Given the stochastic nature of real-world robotics, this sample size is too small to distinguish between genuine generalization and random variance. Increase real-world evaluation to at least 30 episodes per task or provide a statistical test comparing simulation vs. real-world performance distributions.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in the results section requires significant strengthening to support the paper's claims of superiority and robustness. While the experimental design (15 episodes per task) is reasonable for robotics benchmarks, the presentation of results relies exclusively on point estimates (success rates) without any measure of uncertainty.

In Table 1 (sec/04_results.tex), success rates are reported to one decimal place (e.g., 75.6% vs 70.2%). Without confidence intervals (CIs) or standard errors (SE), it is impossible to determine if the observed differences between Guava-Agent-4B and baselines like GPT-5.4 or CaP-Agent0 are statistically significant or merely due to random variance in the 15 trials. For instance, a 5.4% difference in success rate over 15 trials has a standard error of approximately 3.8% (assuming a binomial distribution), meaning the difference is not statistically significant at the 95% level. The authors should calculate and report 95% CIs (e.g., using the Wilson score interval) for all success rates in Table 1 and Figure 2.

Furthermore, the ablation study in Figure 1 (sec/03_method.tex) claims that the "multimodal setting in an iterative workflow demonstrate a consistent higher performance." However, the figure descriptions and text do not mention the number of trials per configuration or include error bars. To validate the claim of "consistent" performance, the authors must show that the performance gap exceeds the variance of the measurements. A simple visual inspection of means is insufficient for scientific rigor in stochastic environments.

Finally, the comparison between SFT and RL post-training (Figure 3, sec/04_results.tex) shows dramatic improvements (e.g., from 0.0% to 93.3% on "place all red objects in basket"). While the magnitude is large, the authors should explicitly state the statistical test used to confirm these gains are not artifacts of the specific random seeds or task instances used. Given the small sample size (15 episodes), non-parametric tests or bootstrapping are recommended over parametric assumptions. The current lack of uncertainty quantification weakens the evidence for the proposed method's effectiveness.

Action items.

- [science] Table 1 and Figure 2 report success rates over 15 and 10 episodes respectively without confidence intervals or standard errors. Given the stochastic nature of embodied manipulation, report 95% CIs or SEs for all quantitative results to assess statistical significance of the reported improvements.
- [science] The claim that Guava-Agent-4B outperforms GPT-5.4 (75.6% vs 70.2%) lacks statistical validation. Perform a paired statistical test (e.g., McNemar's test or bootstrap) across the 15 episodes to determine if the difference is statistically significant ($p < 0.05$) rather than relying on point estimates.
- [science] The ablation study in Figure 1 compares different harness configurations but

does not specify the number of trials per configuration or include error bars. Ensure all ablation results include variance estimates (e.g., error bars representing SE) to support the claim of 'consistent higher performance'.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript is generally well-written, with a clear narrative flow and professional tone suitable for a top-tier venue. The abstract and introduction effectively set up the problem and the proposed solution. However, there are several minor but noticeable errors in spelling, grammar, and consistency that detract from the overall polish of the paper.

Most notably, the framework name "Guava" is misspelled as "Gauva" in the title of Section 3 ("Gauva: Harnessing VLM for Embodied Manipulation"). This is a significant oversight given that the name is central to the paper's identity. Additionally, the section label in the Related Work section is typoed as `sec:realted_word`, which, while a technical detail, suggests a lack of careful proofreading.

There are also minor grammatical slips. In the Results section, a missing parenthesis disrupts the readability of a sentence describing performance metrics ("shell game 6.7%)"). In the Appendix, the phrase "also additionally" is redundant. While these do not impede the understanding of the scientific content, correcting them is essential for a final publication-ready version. The authors should perform a thorough proofread to catch these and any similar minor errors before resubmission.

Action items.

- [writing] In Section 3 (Method), the title misspells the framework name as 'Gauva' instead of 'Guava'. This inconsistency appears in the section header and should be corrected to match the title and abstract.
- [writing] In Section 2 (Related Work), the label for the section is misspelled as 'sec:realted_word' instead of 'sec:related_work'. While this is a LaTeX label, it reflects a proofreading oversight that should be fixed for consistency.
- [writing] In Section 4 (Results), under 'Finding 3', the sentence 'While the SFT policy struggles on both shell game 6.7%) and...' is missing an opening parenthesis before the percentage. It should read '...struggles on both shell game (6.7%) and...'.
- [writing] In the Appendix, under 'Recovery Behavior Generation', the phrase 'We also additionally generate trajectories' contains a redundant adverb. 'Additionally' or 'also' should be removed to improve conciseness.